

Fig. 1: Prototype of short-circuit protection in DC low-voltage systems

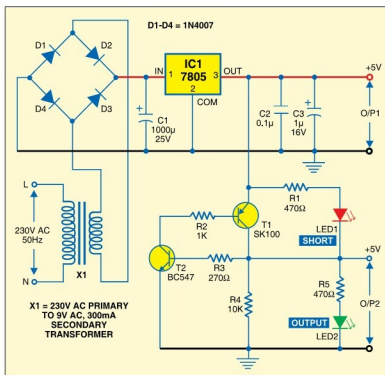


Fig. 2: Circuit diagram of short-circuit protection

cuit, 230V AC is stepped down by transformer X1 (230V AC primary to 0-9V, 300mA secondary), rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 to give regulated 5V (O/P1).

Transistors SK100 and BC547 are used to derive the secondary output of around 5V (O/P2) from the main 5V supply (O/P1). Working of the circuit is simple. When the 5V DC output from regulator IC 7805 is available, transistor BC547 conducts through resistors R1 and R3 and LED1. As a result, transistor SK100 conducts and short-circuit protected 5V DC output appears across

O/P2 terminals. The green LED (LED2) glows to indicate the same, while the red LED (LED1) remains off due to the presence of the same voltage at both of its ends.

When O/P2 terminals short, BC547 cuts off due to grounding of its base. As a result, SK100 is also cut-off. Thus during short-circuit, the green LED (LED2) turns off and the red LED (LED1) glows. Capacitors C2 and C3 across the main 5V output (O/P1) absorb the voltage fluctuations occurring due to short-circuit in O/P2, ensuring disturbance-free O/P1. The design of the circuit is based on the relationship given below:

$$R_b = (H_{FE} \times V_s) / (1.3 \times I_L)$$

where,

R_b = Base resistances of transistors of SK100 and BC547

H_{FE} = 200 for SK100 and 350 for BC547

Switching Voltage V_s = 5V

1.3 = Safety factor

I_L = Collector-emitter current of transistors

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Connect O/P1 and O/P2 terminals on the front panel of the cabinet. Also connect the mains power cord to feed 230V AC to the transformer. Connect LED1 and LED2 for visual indication. ●

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